

System Cabinet – MGD Chassis (Front view)

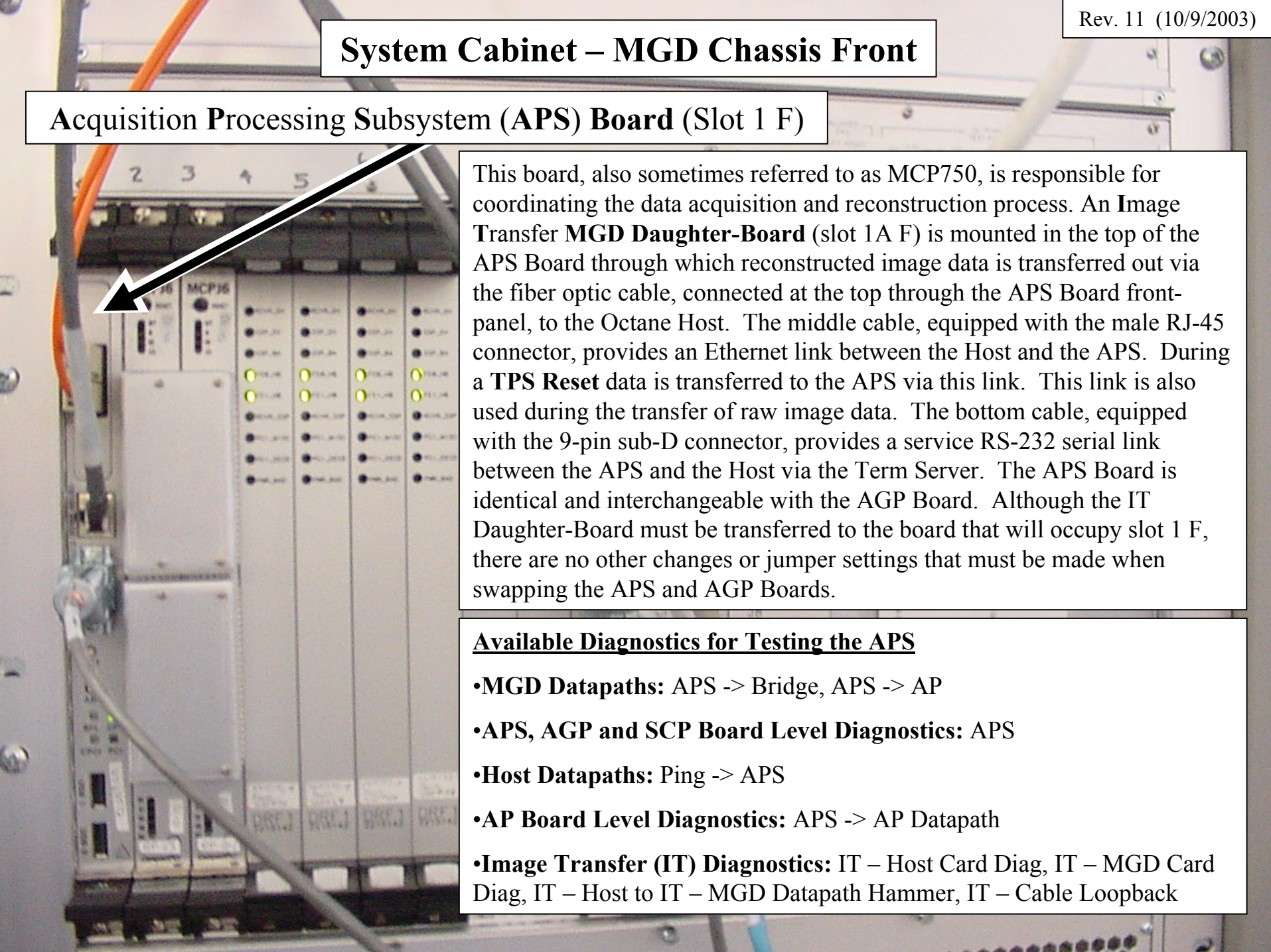
All the circuit boards that comprise the MGD are contained in the **Multi-Generational Data** acquisition Chassis. Boards are located on the front and the rear of the chassis and all the boards interface to a common mid-plane. The mid-plane contains two independent compact-PCI buses that are linked together by the Bridge Board. The compact-PCI bus standard is connectors with male pins present on the mid-plane. Exercise care when replacing or installing circuit boards in the MGD Chassis in order to avoid bending one of these pins. *Before installing or replacing a board into an empty slot, visually inspect the connectors on the mid-plane for bent pins. Also, check the female connector on the board for signs of damage before attempting to seat the board into the slot.* All the slots are keyed in order to make it difficult to put the wrong board into the wrong slot. Two **fans** are located behind the metal cover plate at the top-front of the MGD Chassis. The MGD Chassis contains two independent **temperature sensors**. One is designed to report an over-temp condition when the temperature in the chassis reaches 40 degrees C (104 degrees F) while the other will report an over-temp condition when the temperature in the chassis reaches 50 degrees C (122 degrees F).

The **minimum configuration** of boards installed in the MGD Chassis necessary for applications to successfully load are as follows: APS Board, AGP Board, SCP Board, PS1 and PS2 power supplies, and Bridge Board. All these boards must be installed in the correct slots with all the necessary cabling attached. As long as one is confident that the software is in good working order, failure of the applications software to boot successfully at this point may indicate that one of these boards has failed. Use “cu” (serial connection) or “rlogin” (Ethernet) to login to each the boards and monitor them for suspicious failures during the next boot.

Note the gray tabs inset into the black tabs on the DRF1 Boards, IRF Board, SRF/TRF Board, IRF I/O Board (not shown), and STIF Board (not shown). **In order to remove these boards, the two screws that secure the boards to the MGD Chassis must be retracted and the gray tabs inset in each of the upper and lower black tabs must be pressed inward to allow the black tabs to release.** This must be done before attempting to use the tabs to extract the boards. Failure to release the black tabs first will result in physical damage to the black tabs.

System Cabinet – MGD Chassis Front

Acquisition Processing Subsystem (APS) Board (Slot 1 F)



This board, also sometimes referred to as MCP750, is responsible for coordinating the data acquisition and reconstruction process. An **Image Transfer MGD Daughter-Board** (slot 1A F) is mounted in the top of the APS Board through which reconstructed image data is transferred out via the fiber optic cable, connected at the top through the APS Board front-panel, to the Octane Host. The middle cable, equipped with the male RJ-45 connector, provides an Ethernet link between the Host and the APS. During a **TPS Reset** data is transferred to the APS via this link. This link is also used during the transfer of raw image data. The bottom cable, equipped with the 9-pin sub-D connector, provides a service RS-232 serial link between the APS and the Host via the Term Server. The APS Board is identical and interchangeable with the AGP Board. Although the IT Daughter-Board must be transferred to the board that will occupy slot 1 F, there are no other changes or jumper settings that must be made when swapping the APS and AGP Boards.

Available Diagnostics for Testing the APS

- **MGD Datapaths:** APS -> Bridge, APS -> AP
- **APS, AGP and SCP Board Level Diagnostics:** APS
- **Host Datapaths:** Ping -> APS
- **AP Board Level Diagnostics:** APS -> AP Datapath
- **Image Transfer (IT) Diagnostics:** IT – Host Card Diag, IT – MGD Card Diag, IT – Host to IT – MGD Datapath Hammer, IT – Cable Loopback

System Cabinet – MGD Chassis Front

Array Processor (AP) Board(s) (Slot(s) 2(3) F)



The boards in these slots are responsible for reconstructing an image from raw image data. Two types of AP board are available. The standard type, shown in the picture on the left, has ribbon cable ports on the front of the board. The other type does not have any ribbon cable ports on the front and is explained in more detail in the next slide. The two types of AP Board cannot be used together. The MGD Chassis AP Board(s) can be configured in one of 2 ways. **Reflex 200** is the basic, standard configuration in which one AP Board equipped with two processor daughter cards is installed in slot 2, F while slot 3, F is left empty. **Reflex 400**, which provides the highest level of performance, describes the third configuration in which two AP Boards (the same type) equipped with two processor daughter cards each are installed in slots 2 F and 3 F as shown in the background illustration. The “Master” AP Board is always located in slot 2 while the “Slave” AP Board is always located in slot 3. “Master” and “Slave” board designations are determined by jumper settings on the standard AP board. See the appropriate AP Board replacement documentation for details concerning “Master” and “Slave” jumper or switch configuration for the boards.

Available Diagnostics for Testing the AP Board(s)

- MGD Board Level Diagnostics:** AP Recon
- AP Board Level Diagnostics:** APS -> AP Datapath, AP Recon Test, Single Board AP Mem (for MGD with Reflex 200), Dual Board AP Mem (for MGD with Reflex 400)
- AP Functional Check:** Checks MGD Chassis AP Board(s) configuration

System Cabinet – MGD Chassis Front

Array Processor (AP) Board(s) (Slot(s) 2(3) F)



Close-up of installed PIM connector.

The AP boards in these slots do not have ribbon cable ports. These do offer cost advantages over the boards discussed in the previous slide but do not offer any performance advantages. Two types of AP board are available. The two types of AP Board should not be used together. **Reflex 200B** is the basic, standard board configuration in which slot 2, F and slot 3, F are equipped with the Reflex 200B boards in a Master/Slave configuration. **Reflex 400B**, which provides the highest level of performance, is the optional configuration in which slot 2, F and slot 3, F are equipped with the Reflex 400B boards in a Master/Slave configuration. The “Master” AP Board is always located in slot 2 while the “Slave” AP Board is always located in slot 3. “Master” and “Slave” board designations are determined by a single rotary switch on each board. Since the boards are not interconnected with ribbon cables it is necessary to install a PIM connector across slots 2, R and 3, R on the rear of the MGD Chassis. The PIM connector accomplishes the same function as the ribbon cables. See the appropriate AP Board replacement documentation for details concerning “Master” and “Slave” switch configurations for each board and PIM installation.

Available Diagnostics for Testing the AP Board(s)

- **MGD Board Level Diagnostics:** AP Recon
- **AP Board Level Diagnostics:** APS -> AP Datapath, AP Recon Test, Single Board AP Mem (for MGD with Reflex 200B), Dual Board AP Mem (for MGD with Reflex 400B)
- **AP Functional Check:** Checks MGD Chassis AP Board(s) configuration

System Cabinet – MGD Chassis Front

Digital Receive Filter 1 (DRF1) Board (Slots (4),(5),6,7 F)

These boards support 2 receive channels each at +/- 250KHz bandwidth. The standard configuration is for each MGD Chassis to contain 2 of these boards, one in slot 6 (DRF 2; channels 3 & 4) and one in slot 7 (DRF 1; channels 1 & 2), and thus support a minimum of 4 receive channels. Two optional DRF1 Boards can be installed, one in slot 4 (DRF 4; channels 7 & 8) and one in slot 5 (DRF 3; channels 5 & 6), to support a total of 8 receive channels and bring the total number of boards to 4 (as shown in the illustration). It is important to understand that the DRF1 designations are not slot or board dependant but are automatically assigned from the right-most slot to the left based on the number of boards in the MGD Chassis. For example, if slots 6 & 7 were empty (not a normal configuration) but slots 4 & 5 were filled then DRF 1 would be the board in slot 5 and DRF 2 would be the board in slot 4. The data handled by these boards, as is true for all the boards in the MGD Chassis, is digital and not analog data. The function of these boards is to provide receiver channel pre-reconstruction processing. After a successful **TPS Reset** the LEDs on the front of the DRF1 Boards should flash sequentially from the right-most DRF1 Board (first to be initialized) to the left (last to initialize).

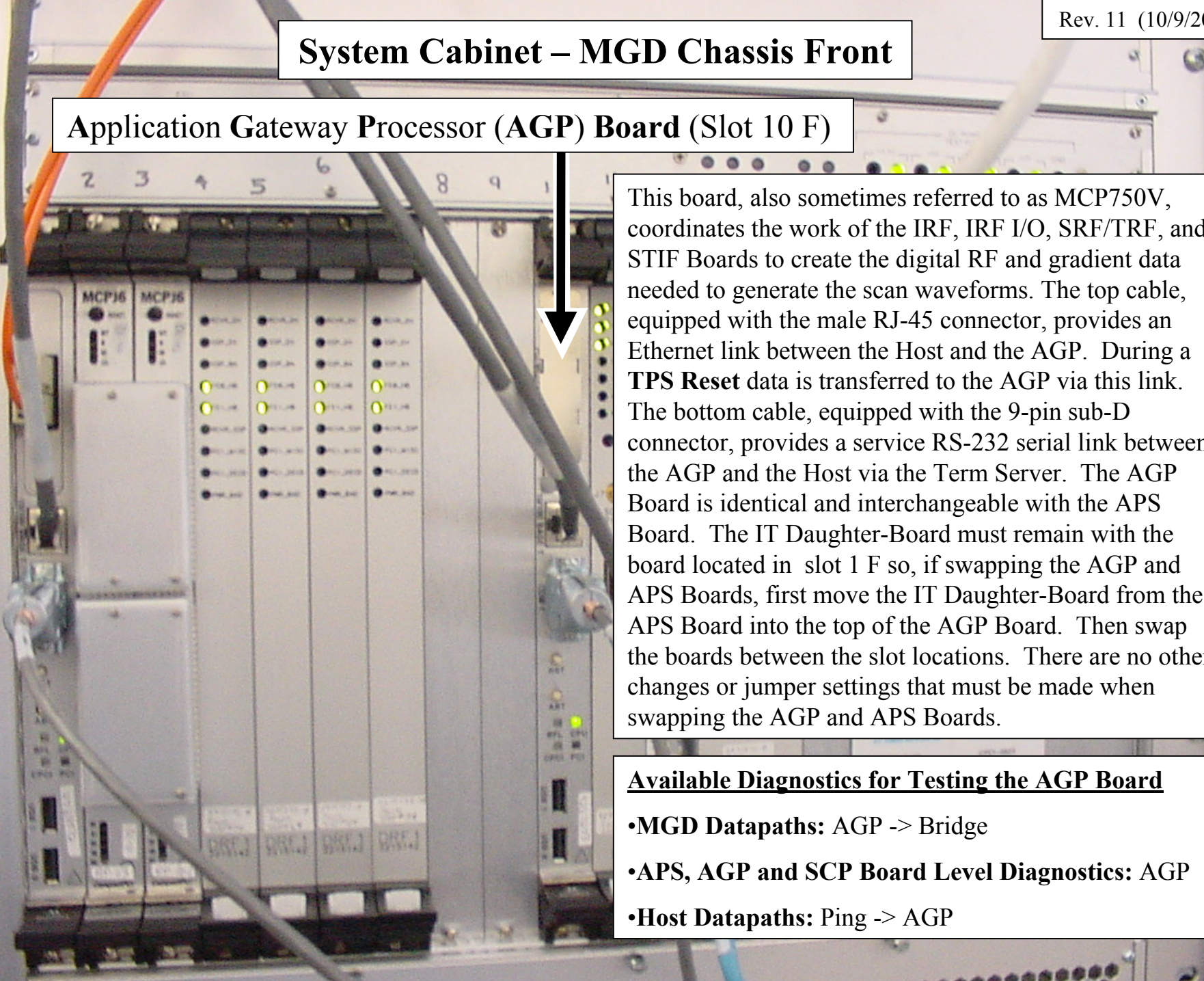
Optional

Available Diagnostics for Testing the DRF1 Boards

- MGD Board Level Diagnostics:** DRF 1 BLD (for single DRF1 board or, if multiple DRF1 boards, board in right-most slot), DRF 2 BLD (for board left of DRF 1), DRF 3 BLD (for board left of DRF 2), DRF 4 BLD (for board left of DRF 3; slot 4)
- MGD Host Datapaths:** IRF -> DRF (all), SRF -> DRF (all)

System Cabinet – MGD Chassis Front

Application Gateway Processor (AGP) Board (Slot 10 F)



This board, also sometimes referred to as MCP750V, coordinates the work of the IRF, IRF I/O, SRF/TRF, and STIF Boards to create the digital RF and gradient data needed to generate the scan waveforms. The top cable, equipped with the male RJ-45 connector, provides an Ethernet link between the Host and the AGP. During a **TPS Reset** data is transferred to the AGP via this link. The bottom cable, equipped with the 9-pin sub-D connector, provides a service RS-232 serial link between the AGP and the Host via the Term Server. The AGP Board is identical and interchangeable with the APS Board. The IT Daughter-Board must remain with the board located in slot 1 F so, if swapping the AGP and APS Boards, first move the IT Daughter-Board from the APS Board into the top of the AGP Board. Then swap the boards between the slot locations. There are no other changes or jumper settings that must be made when swapping the AGP and APS Boards.

Available Diagnostics for Testing the AGP Board

- MGD Datapaths:** AGP -> Bridge
- APS, AGP and SCP Board Level Diagnostics:** AGP
- Host Datapaths:** Ping -> AGP

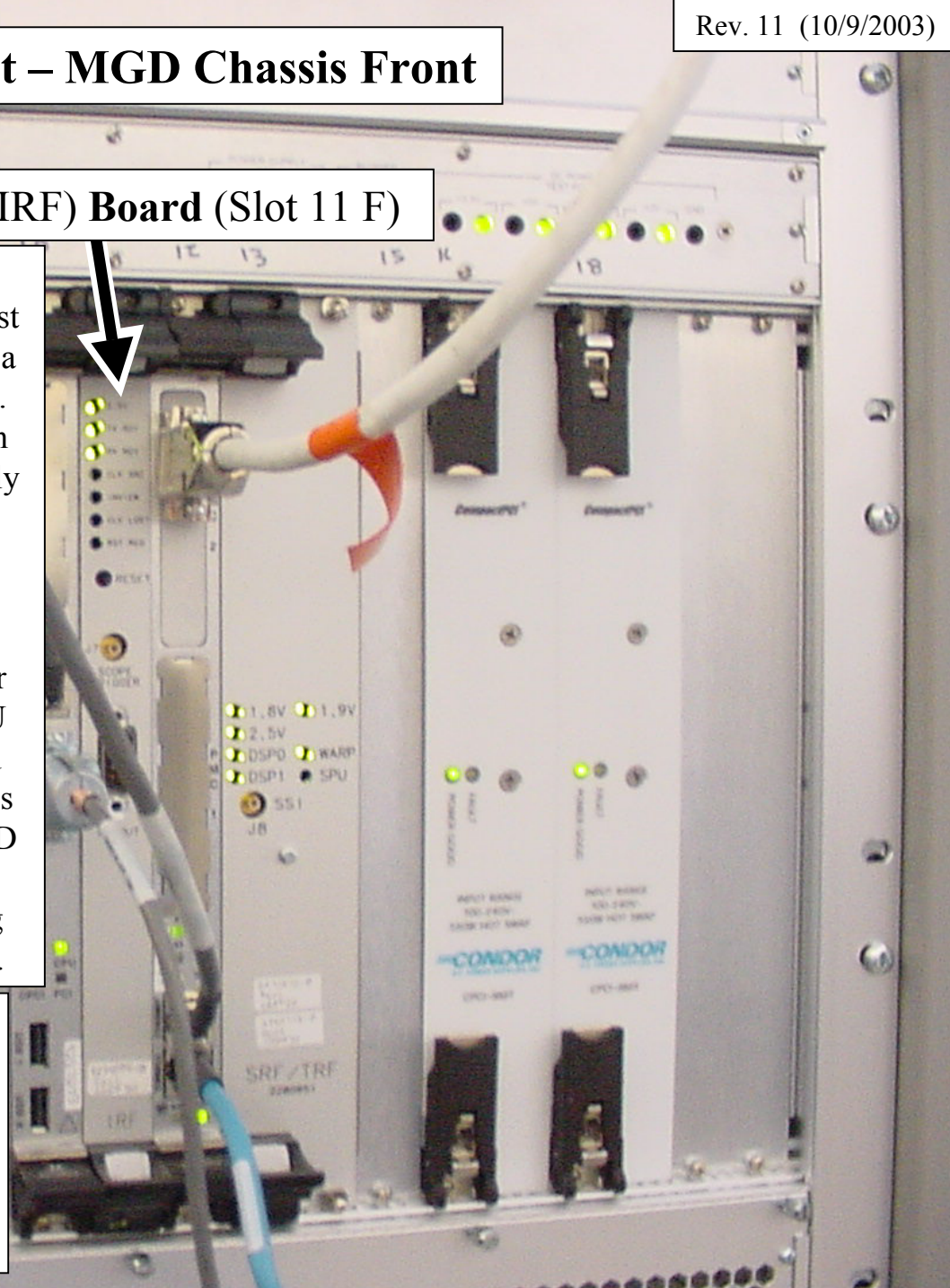
System Cabinet – MGD Chassis Front

Interface & Remote Functions (IRF) Board (Slot 11 F)

The IRF Board has two main functions. The first is to provide a master data acquisition clock signal to the rest of the MGD components. This signal is obtained from a 20MHz reference provided by the RRF Master Exciter. The second main function of the IRF Board is to obtain digitized data from the SRF/TRF Board and then supply this data to IRF I/O for transfer to the Exciter(s) in the RRF so that the RF waveforms can be generated. FRESBECC B0 compensation data is also supplied from this board to the IRF I/O for transfer to the Exciter(s) in the RRF. The IRF Board also provides for control and status monitoring of coil temperature, PDU mode, scan room door status, bore fan, bore lights, and alignment lights. The RESET button on the front of this board can be used to initiate a manual reset of the MGD if the serial reset line is not working. Press this button and then either TPS Reset or boot applications. Timing is critical so if it does not work the first time, try again.

Available Diagnostics for Testing the IRF Board

- MGD Board Level Diagnostics:** IRF BLD
- MGD Datapaths:** IRF -> DRF, IRF -> IRF I/O, SRF -> IRF
- RRF Datapath** IRF -> RRF



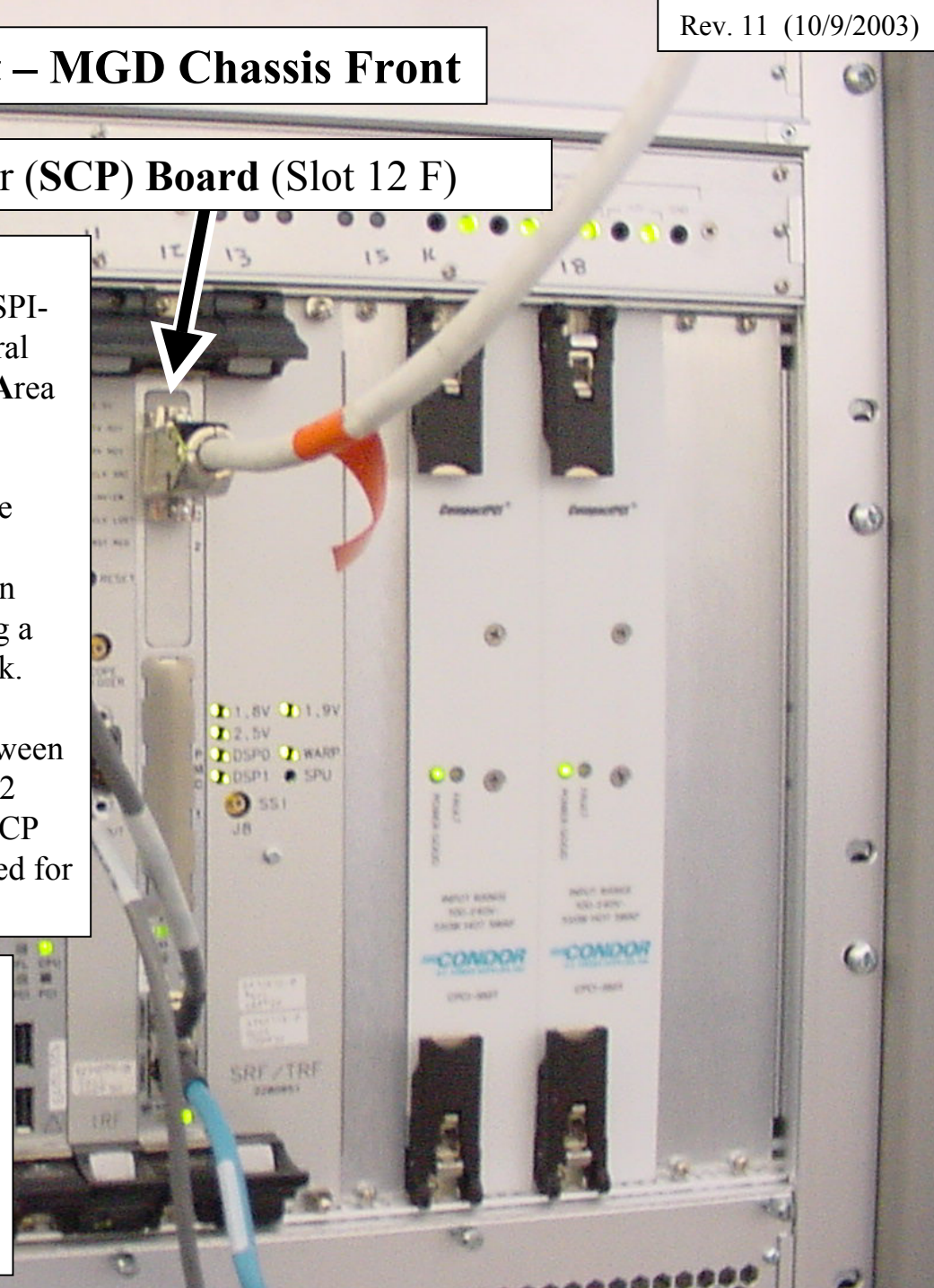
System Cabinet – MGD Chassis Front

Scan Control Processor (SCP) Board (Slot 12 F)

This board is responsible for providing scan state information to the APS and AGP. It also performs SPI-type functions by managing the MDS serial peripheral devices, the SRI, and the CAN Link. A **Controller Area Network (CAN) Daughter-Board** (slot 12A F) is mounted in the top of the SCP Board through which communication is managed between the SCP and the RRF and Driver Module. The middle gray cable, equipped with the male RJ-45 connector, provides an Ethernet link between the Host and the SCP. During a **TPS Reset** data is transferred to the SCP via this link. The bottom blue cable, equipped with an RJ-45 connector, provides a service RS-232 serial link between the SCP and the Host via the Term Server. The SCP2 Board, while functionally equivalent to a standard SCP Board, is equipped with more memory and is required for certain purchased scan options.

Available Diagnostics for Testing the SCP

- **APS, AGP and SCP Board Level Diagnostics:** SCP
- **SRI Datapaths:** SCP Fiber Loopback, SCP STIF Loopback
- **MGD Host Datapaths:** Ping -> SCP



System Cabinet – MGD Chassis Front

Sequence Related Functions / Trigger & Rotational Functions (SRF/TRF) Board (Slots 13,14 F)

This two-board assembly generates the digital data needed for producing the RF and gradient waveforms necessary for scanning. It also provides these functions:

- Oblique Plane Imaging rotation algorithm (WARP/TRF)
- Real-time eddy current compensation (WARP/TRF)
- Scan sequence control (SRF)
- Physiological gating algorithms (SPU/TRF)
- Physiological waveform display generation (SPU/TRF)

The functions of **S**ignal **P**rocessing **U**nit for physiological gating (**SPU**) and **W**aveform and **R**otation **P**rocessor (**WARP**), as listed above, are done on this board. This board communicates with the PAC and gradients via high-speed serial interfaces.

After a successful **TPS Reset** the DSP0, DSP1, and WARP LEDS should be flashing.



Available Diagnostics for Testing the SRF/TRF

- **MGD Board Level Diagnostics:** TRF BLD, SRF BLD
- **MGD Datapaths:** SRF -> DRF, SRF -> IRF, SRF -> TRF

System Cabinet – MGD Chassis Front

Power Supply Modules (PS1 F, PS2 F)

Both modules work in tandem to supply power to the MGD Chassis. The output voltages these boards provide cannot be adjusted. During normal operation the green POWER SUPPLY LED will be illuminated. Some types of power supply failures will cause the red FAULT LED to illuminate. A row of Green indicator LEDs located above the supplies in the MGD Chassis provide a visual indication of each supply voltage. Test points next to each LED provide a convenient measurement location for checking the condition of each supply voltage with a Digital Multimeter. Please note that the MGD hardware does **NOT** support “hot swapping” (i.e. removing and installing power supply with power on) of power supplies. In the event that one of the modules fails, it is recommended to not operate the MGD Chassis with a single supply. One reason for not doing this is that the air flow will be compromised due to the missing supply. This forces the other boards to operate at less than optimal temperatures which could cause damage. Another reason for not doing this is that the remaining power supply is operating over capacity to supply all the MGD Chassis power needs. This can damage the power supply and introduce intermittent failures that may be difficult to isolate. The more options installed (4 DRF1 Boards, Reflex 400(B)), the more stress and possible damage a single supply may encounter.



System Cabinet – MGD Chassis Rear

This board provides a communication bridge between the AGP and APS sides of the two PCI buses. The two LEDs on the rear of the board provide a visual indication of the status of the bus power supplies.

Bridge Board (Slot 8 R)

Available Diagnostics for Testing the Bridge Board

•**MGD Host Datapaths:** Except for the Ping -> SCP test, the other tests indirectly test the functionality of the Bridge by checking the identity of the APS and AGP Boards. These boards will assume default identities (MCP750 for the APS and MCP750V for the AGP) if the boards cannot communicate because the Bridge Board has failed. The assumption of default identities by the APS and AGP Boards does not always indicate that the Bridge Board has failed. It only indicates that during the initialization period the AGP and APS boards could not, for some reason, communicate.

•**MGD Datapaths:** APS -> Bridge, AGP -> Bridge

System Cabinet – MGD Chassis Rear

Interface & Remote Functions Input/Output (IRF I/O) Board (Slot 11 R)

This board is a communications interface between the IRF Board and RRF Module and IRF Board and system control and status functions. The fiber optic cable interfaces to the “Gigalink” port on the rear of the board and connects to the RF-DIF Board in the RRF Module. This board also provides a **System Monitor and Control (SMC)** interface by which the IRF Board can monitor the PDU mode, Gradient Coil Temperature, Scan Room Door status, and E-Stop signal and control the Alignment Lights, Bore Lights, and Bore Fan.

Available Diagnostics for Testing the IRF I/O Board

- MGD Datapaths:** IRF -> IRF I/O
- RRF Datapaths:** IRF -> RRF (indirectly tested)

System Cabinet – MGD Chassis Rear

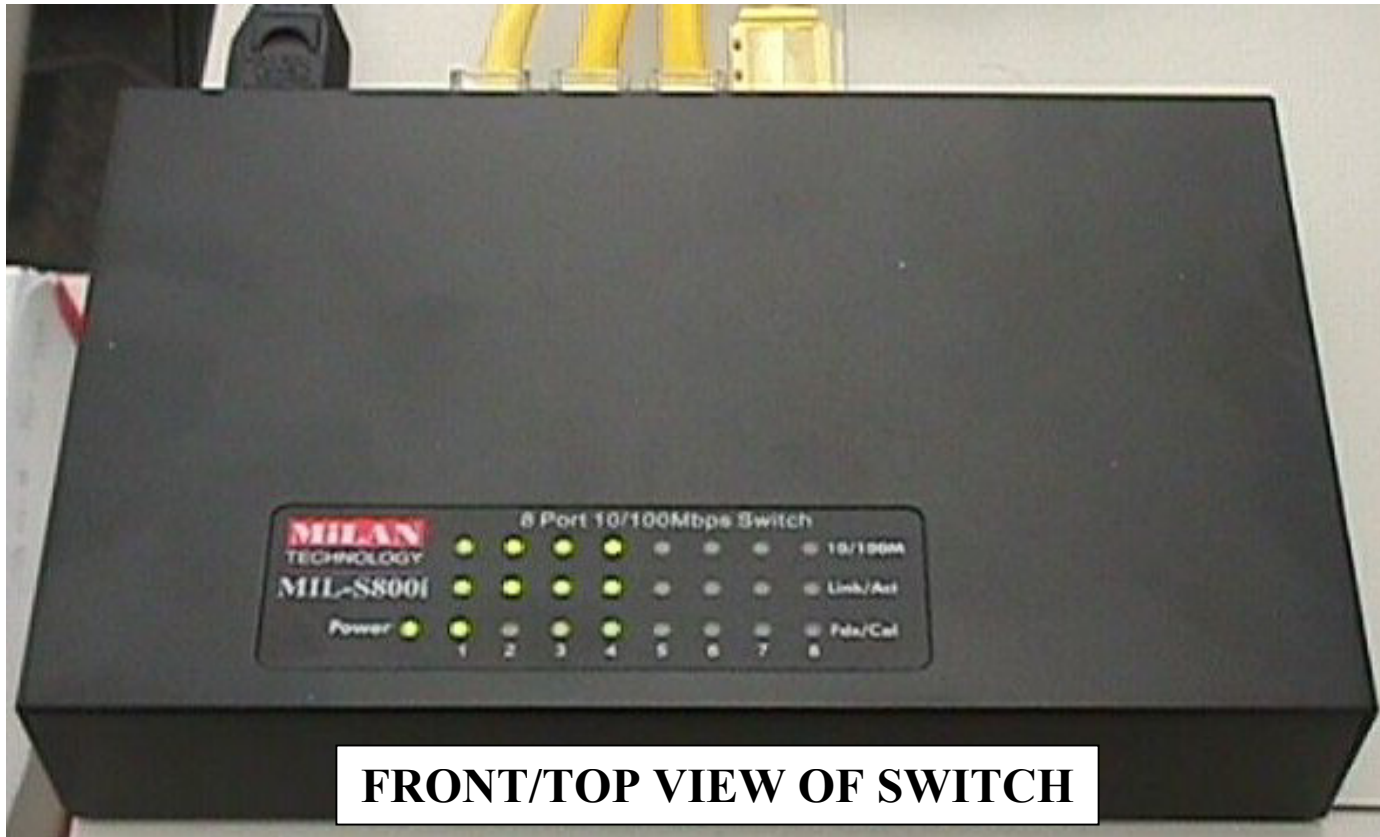
SRF / TRF InterFace (STIF) Board (Slots 13,14 R)

This board provides the fiber optic interfaces (fiber cables are not shown connected) for the fiber cables connecting to the **Gradient Processor (GP) Board** inside the **ACGD Cabinet**, the **SRI** inside the scan room, the **PAC Module** inside the scan room, and the **MDS Link** that interconnects the System Cabinet to the SRF and ACGD Cabinets. The coaxial cable connecting to J5 provides the 60Hz line clock signal from the AC distribution panel inside the cabinet. The gray cable shown in the illustration equipped with the 9-pin sub-D connector provides an RS-232 serial link between the STIF Board and the Host via the Term Server. The function of this link is to supply the signal necessary to start a **TPS Reset** when this is selected from the Operators Workstation. Physiological waveform data, physiological control data, and hardkey control data is transferred from connector J3 (cable not shown connected).

Available Diagnostics for Testing the STIF Board

- SRI Datapaths:** SCP STIF Loopback
- Gradient Diagnostics:** MDS STIF Loopback Test
- MDS Diagnostics:** MDS STIF Loopback Test
- SRI Functional Tests:** STIF – SRI Fiber Optic

System Cabinet – Network Switch

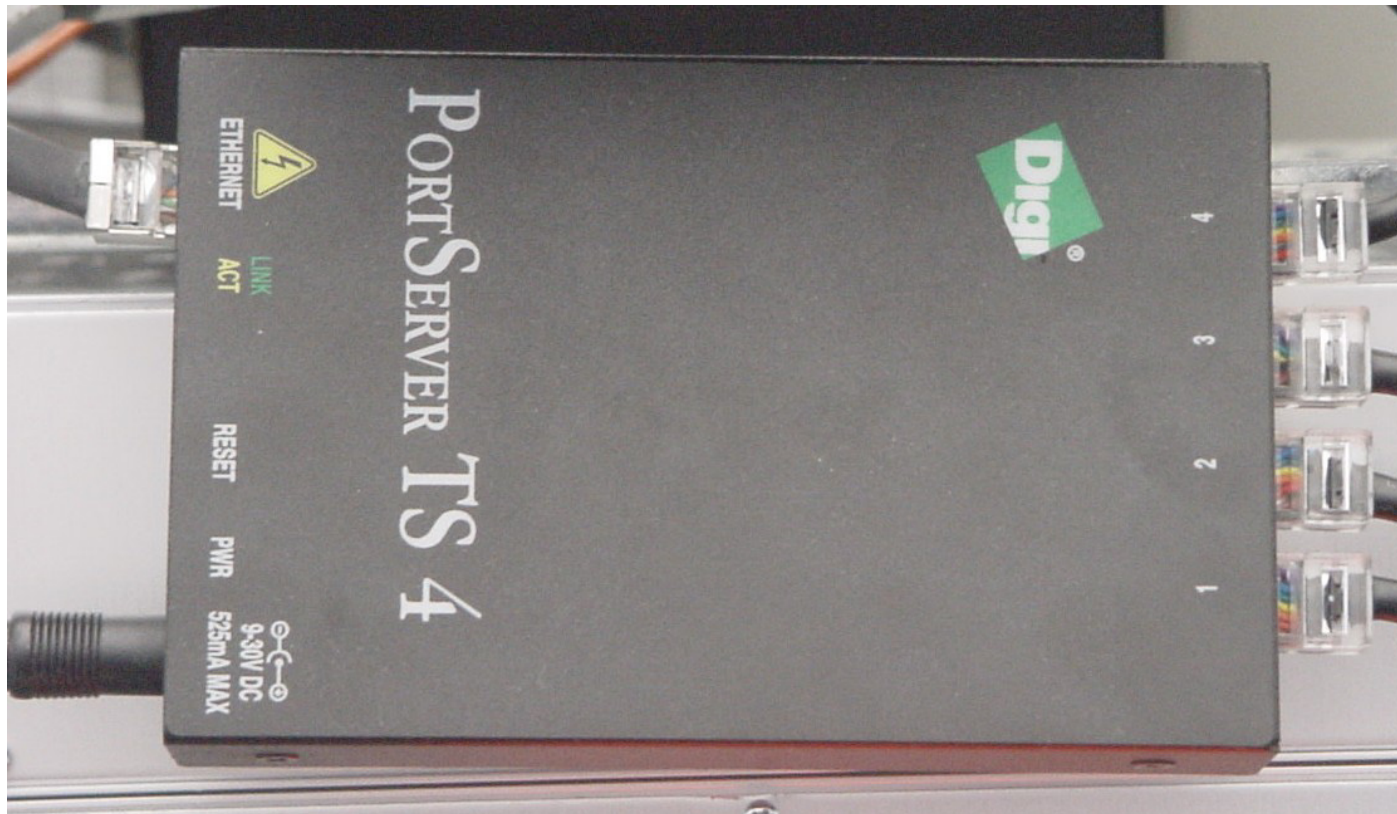


FRONT/TOP VIEW OF SWITCH

The Network Switch is secured by a removable Velcro strap to a metal plate situated above the MGD Chassis. It provides 8 ports but only 5 are currently used. The status of each port is indicated by three sets of green LEDs. Refer to the Illustration above. These indicators can be viewed by removing the Velcro strap from the Switch and then angling the rear of the unit away from the metal plate and up into a vertical position. The status of each row of LEDs is as follows: **10/100M** – Lit LED indicates link at 100Mbps and unlit indicates 10Mbps or no link.

Link/Act – Lit LED indicates good connection, flashing indicates port is transferring data, and unlit indicates that the port has no link. **Fdx/Col** – Lit LED indicates full-duplex mode, unlit indicates half-duplex mode, and flashing indicates data collision. In normal operation 5 columns of LEDs and Power LED should be lit/active. The Ethernet lines can be connected to ports 1 – 8 in any order. Connections can be swapped between any of ports 1 – 8 for troubleshooting.

System Cabinet – Term Server



The Term Server permits RS-232C serial connection to each of the processor boards and the STIF Board by utilizing the existing Ethernet link. Serial ports are selected at the Term Server based on the information received in the Ethernet signal. The serial lines on the Term Server are connected as follows: Port 1 connects to the APS, Port 2 connects to the AGP, Port 3 connects to the SCP, and Port 4 connects to the Host serial reset port on the STIF Board. With the exception of the Port 4 connection to the STIF Board, it is not necessary for any of the other serial ports to be functional in order for the system to boot or for the MGD to reset. If the Port 4 serial connection is not functioning, it may not be possible to complete a TPS Reset from the Host. The Ethernet cable from the Term Server can connect to any open port on the Ethernet Switch. The Term Server is physically located next to the Ethernet Switch.